

TL072 Diode Fuzz

Build a single-supply op-amp gain stage with feedback diodes and a low-pass tone network.

EVIDENCE STATUS

Calculated circuit relationships + browser-simulated audio. Construction layout is diagram-audited but NOT physically tested.

1. Project purpose

Build a single-supply op-amp gain stage with feedback diodes and a low-pass tone network.

How it works

The op amp amplifies around a buffered 4.5 V reference. Antiparallel silicon diodes conduct on opposite waveform halves and limit feedback voltage, creating soft clipping and added harmonics.

Calculated relationship

Formula: Inverting gain magnitude before clipping: $|A| = R_f/R_{in}$; tone corner: $f_c = 1/(2 \pi R_{tone} C_{tone})$

Substitution: $R_f = 470 \text{ k}\Omega$, $R_{in} = 10 \text{ k}\Omega$, $R_{tone} = 10 \text{ k}\Omega$, $C_{tone} = 22 \text{ nF}$

Result: Calculated unclipped gain: 47x. Calculated tone corner: 723 Hz. Diodes reduce practical closed-loop gain when conducting.

Safety boundary

Use only a regulated, current-limited 9 V DC supply. Disconnect power before moving parts. Check IC orientation, diode bands, electrolytic polarity, and rail continuity. Connect the output only to a powered speaker, amplifier, or audio-interface input - never directly to a bare low-impedance speaker.

2. Required components

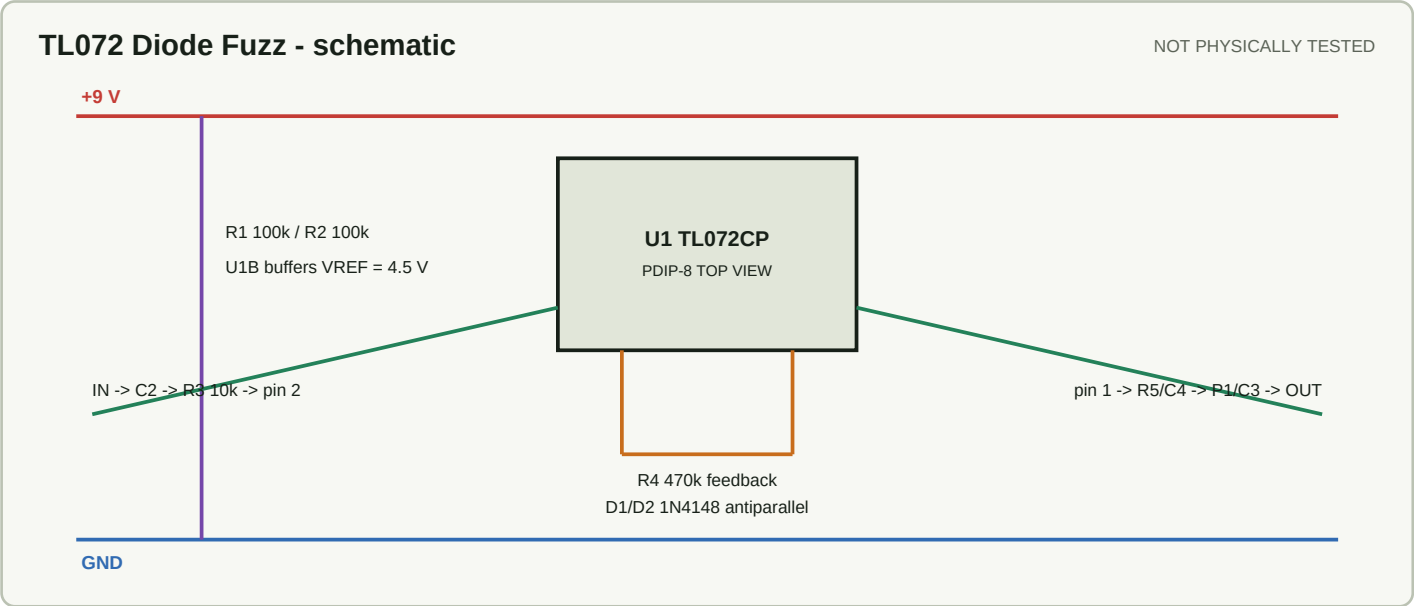
Ref	Exact component / value	Qty
BB1	BB830-style solderless breadboard	1
U1	TI TL072CP PDIP-8	1
R1, R2	100 kOhm 1/4 W bias divider	2
R3	10 kOhm 1/4 W input resistor	1
R4	470 kOhm 1/4 W feedback resistor	1
D1, D2	onsemi 1N4148 DO-35 diode	2
R5	10 kOhm 1/4 W tone resistor	1
C4	22 nF film tone capacitor	1
C1	10 uF 25 V electrolytic	1
C2	100 nF film input capacitor	1
C3	1 uF 25 V electrolytic	1
C5	100 nF ceramic bypass	1
P1	Bourns PTV09 100 kOhm linear	1
J1, J2	6.35 mm mono jack	2
PS1	Regulated current-limited 9 V DC supply	1

Pinout verification

TL072CP package view: TOP VIEW. In the supplied horizontal breadboard layout the notch faces left; DIP numbering proceeds counterclockwise.

Pin	Name	Role
1	1OUT	Amplifier A output
2	1IN-	Amplifier A inverting input
3	1IN+	Amplifier A non-inverting input
4	VCC-	Ground in this build
5	2IN+	Amplifier B non-inverting input
6	2IN-	Amplifier B inverting input
7	2OUT	Amplifier B output
8	VCC+	+9 V supply

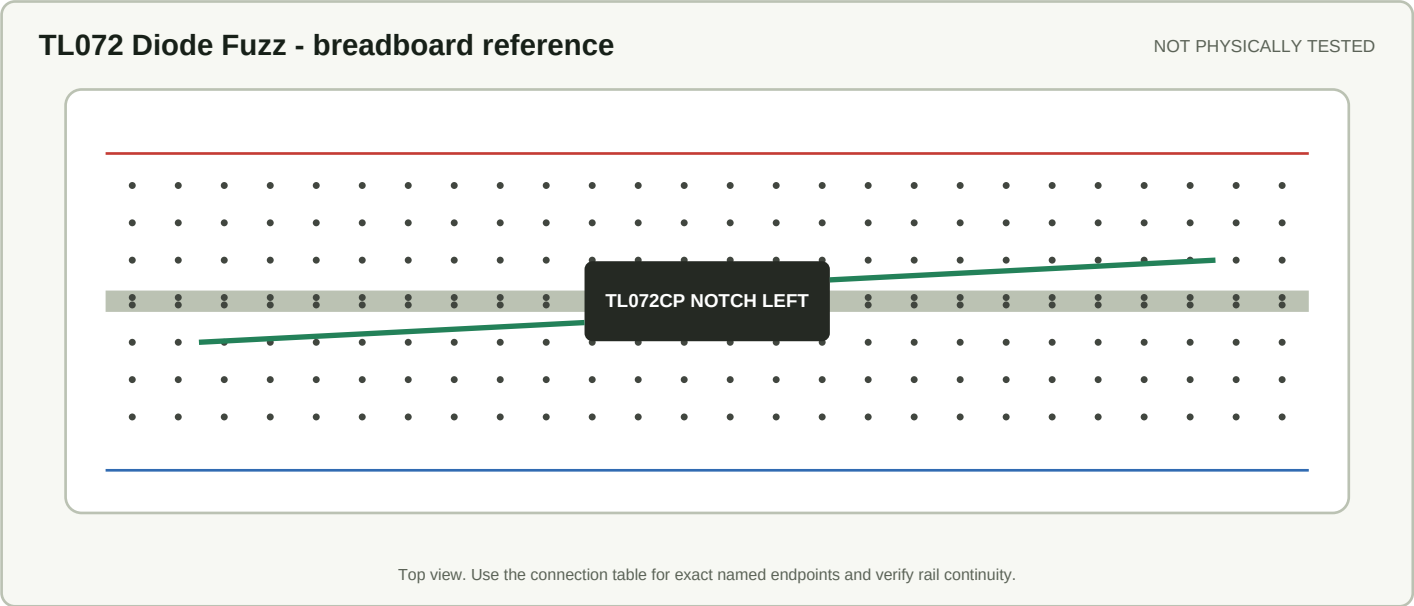
3. Labeled schematic



4. Canonical connections

Net	From	To	Wire
Power	+9 V / ground	U1 pin 8 / pin 4; C5 across supply	red/black
VBIAS	+9 V through R1	junction through R2 and C1 to ground	purple
VREF	VBIAS to U1 pin 5	U1 pins 6 and 7 joined; pin 7 to pin 3	purple
Input	J1 tip through C2 and R3	U1 pin 2; J1 sleeve to ground	blue
Feedback	U1 pin 1 through R4	U1 pin 2	orange
Clip +	D1 anode at pin 1	D1 cathode band at pin 2	yellow
Clip -	D2 anode at pin 2	D2 cathode band at pin 1	yellow
Tone	U1 pin 1 through R5	TONE; C4 from TONE to ground	green
Level	TONE to P1 lug 3	lug 1 ground; wiper through C3 to J2 tip	green
Ground	J1 and J2 sleeves	Shared ground rail	black

5. Breadboard reference



Breadboard orientation

This drawing is a top-view placement reference. The connection table is authoritative for every electrical net. Typical breadboards connect A-E in each numbered row and F-J separately across the center trench; power-rail continuity varies by product and must be measured with power removed.

6. Power-off assembly

- 01** Disconnect power, verify rail continuity, and place U1 across the center trench at columns 12-15 with notch left.
- 02** Connect U1 pin 8 to +9 V and pin 4 to ground. Add C5 close to these supply pins.
- 03** Build VBIAS: R1 from +9 V to the junction, R2 from junction to ground, C1 positive to junction and negative to ground.
- 04** Connect VBIAS to pin 5; join pins 6 and 7; connect pin 7 to pin 3. Pin 7 is buffered VREF.
- 05** Connect J1 tip through C2 and R3 to pin 2. Connect J1 sleeve to ground.
- 06** Connect R4 between pins 1 and 2. Add D1 and D2 antiparallel across R4; inspect both cathode bands.
- 07** Connect pin 1 through R5 to TONE and C4 from TONE to ground.
- 08** Connect TONE to P1 lug 3, lug 1 to ground, and the wiper through C3 to J2 tip. C3 positive faces P1.
- 09** Set P1 low, inspect polarity and pin orientation, then apply current-limited 9 V.

7. Test procedure and expected results

Start with the supply current limit low and the output level down. Use a multimeter in DC-voltage mode with the black probe on the ground rail. Confirm the power and bias readings before connecting audio. Expected readings below are predictions, not measurements.

Checkpoint	Calculated / expected observation
1	U1 pin 8 to pin 4: about 9 V DC
2	VBIAS and U1 pin 7: about 4.5 V DC
3	U1 pin 1: audio centered near 4.5 V
4	J2 after C3: near 0 V DC with clipped audio

8. Troubleshooting

1. Remove power. Confirm +9 V and ground are not shorted using resistance or continuity mode.
2. Confirm the IC notch direction and locate pin 1 again. A mirrored package is the most serious common wiring error.
3. Check every polarized capacitor and every diode band against the connection table.
4. Measure the supply at the IC pins. If it is missing, repair power wiring before testing signals.
5. Compare one net at a time with the canonical table. Do not change several wires at once.
6. If DC checks pass but sound is missing, trace from output toward input with a powered audio probe or oscilloscope under instructor supervision.
7. Component tolerances change pitch, cutoff, gain, and bias. A difference from the browser model is not automatically a fault.

Sources

TI TL072: <https://www.ti.com/product/TL072>

onsemi 1N4148: <https://www.onsemi.com/pdf/datasheet/1n914-d.pdf>

Bourns PTV09: <https://www.bourns.com/docs/product-datasheets/PTV09.pdf>

Wire audit statement

The connection table, schematic labels, and breadboard IC orientation were reconciled for this guide. Off-board potentiometers and jacks are identified by terminal name in the table. No physical build or measurement was performed; complete a continuity and power-rail audit before first power-up.